

MUIR AAF, PA, USA

WMO: 725144

Lat: 40.433N

Lon: 76.567W

Elev: 149

StdP: 99.55

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 54723

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-10.8	-21.4	0.6	-11.4	-18.9	0.7	-8.7	10.7	0.5	9.2	-2.2	2.9	280	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.9	23.2	31.5	22.5	30.1	21.9	25.0	29.6	24.2	28.6	23.5	27.6	3.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.0	26.4	23.0	18.0	25.8	22.3	17.3	25.2	76.6	29.5	73.4	28.7	70.5	27.3	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.5	-17.0	35.7	3.5	1.9	-19.5	37.1	-21.6	38.3	-23.6	39.3	-26.2	40.8		
			-17.6	26.6	3.2	0.9	-19.9	27.2	-21.8	27.7	-23.6	28.2	-25.9	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-1.9	-0.2	4.7	11.3	16.8	21.2	23.8	22.4	19.0	12.7	6.0	1.5
	DBStd	9.82	5.26	5.25	5.02	4.73	4.26	3.16	2.70	2.63	3.76	4.49	4.37	4.39
	HDD10.0	1295	369	287	176	38	2	0	0	0	0	22	134	266
	HDD18.3	2991	627	518	422	214	83	11	1	2	38	182	371	522
	CDD10.0	1846	1	2	13	78	212	337	427	384	269	107	13	3
	CDD18.3	499	0	0	0	4	34	98	169	128	57	9	0	0
	CDH23.3	4377	0	1	3	91	340	878	1580	979	448	58	0	0
	CDH26.7	1493	0	0	0	24	97	278	636	302	143	12	0	0
Wind	WSAvg	2.2	2.7	2.8	3.0	2.9	2.1	2.0	1.7	1.5	1.6	2.0	2.2	2.5
Precipitation	PrecAvg	1142	81	66	95	91	101	103	113	108	115	101	75	92
	PrecMax	1825	166	141	172	228	170	253	197	206	413	249	135	169
	PrecMin	738	35	14	22	29	29	23	38	34	4	18	21	17
	PrecStd	208	33	33	40	44	38	49	40	42	84	57	31	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	18.3	22.9	29.8	31.6	34.0	35.8	33.1	32.9	28.2	20.6	16.4
		MCWB	12.9	12.1	13.7	17.8	22.1	24.1	23.6	23.1	23.3	21.0	14.9	15.2
	2%	DB	9.8	13.6	19.3	26.0	29.3	31.5	33.7	31.4	30.6	24.7	17.5	12.9
		MCWB	7.6	9.6	12.8	16.7	20.9	22.3	23.7	22.8	22.2	19.4	13.7	11.5
	5%	DB	7.5	10.0	15.8	22.6	27.3	29.7	31.9	29.9	28.0	22.4	15.3	10.2
		MCWB	6.0	6.4	10.7	14.8	19.9	21.5	22.9	22.2	21.1	18.0	11.6	8.5
	10%	DB	5.5	6.8	12.8	20.0	24.9	28.1	30.2	28.4	25.9	20.3	13.0	8.0
		MCWB	3.7	3.9	8.6	13.1	18.6	20.6	22.2	21.5	19.9	16.5	9.7	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	15.6	15.9	21.1	24.5	25.2	26.2	25.7	25.1	22.6	16.8	15.1
		MCDB	14.4	17.7	19.6	24.6	28.2	30.4	32.4	29.7	29.4	25.1	17.9	15.9
	2%	WB	8.3	10.6	14.3	18.3	22.6	24.1	25.1	24.7	23.7	21.3	14.9	12.1
		MCDB	8.7	12.5	18.4	22.9	27.0	29.2	30.2	28.4	27.6	23.6	16.8	12.5
	5%	WB	6.1	6.8	11.7	16.6	21.3	23.1	24.2	23.7	22.6	19.6	12.6	9.0
		MCDB	7.3	9.2	14.8	21.3	25.3	27.5	28.9	27.2	26.0	21.7	14.5	9.6
	10%	WB	3.9	4.4	9.2	14.7	20.2	22.2	23.5	22.8	21.5	17.4	10.5	6.4
		MCDB	5.2	6.3	11.9	18.9	23.3	26.2	28.0	26.2	24.5	19.3	12.2	7.4
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.0	10.0	11.5	11.2	11.4	11.5	10.8	11.2	10.7	9.4	7.5
		MCDBR	10.1	13.4	14.9	16.5	14.9	13.7	13.7	12.6	13.8	13.2	13.1	10.1
	5% WB	MCWBR	8.8	9.9	9.4	8.5	7.4	6.1	5.3	5.4	6.3	7.6	8.8	9.0
		MCWBR	9.3	12.6	12.9	14.0	12.6	11.9	11.4	10.4	11.0	10.4	10.5	8.9
Clear-Sky Solar Irradiance	taub		0.311	0.332	0.366	0.392	0.425	0.493	0.498	0.492	0.411	0.369	0.328	0.309
		taud	2.422	2.336	2.328	2.290	2.228	2.073	2.071	2.076	2.275	2.401	2.480	2.471
	Ebn at Noon		848	877	883	881	856	795	786	779	825	826	823	823
		Edh at Noon	81	102	115	128	139	163	162	157	120	95	77	72
All-Sky Solar Radiation	RadAvg		1.79	2.69	3.61	4.67	5.26	5.82	5.90	5.13	4.20	2.95	2.06	1.50
	RadStd		0.15	0.35	0.27	0.45	0.62	0.49	0.32	0.29	0.56	0.33	0.20	0.18

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118	+64

Nomenclature: See separate page